

1(a)(i)	wind speed ;	1												
1(a)(ii)	<i>any six from:</i> 1 method to use fan(s), to obtain a minimum of five different wind speeds ; 2 <i>ref.</i> to standardised variable for using the fan ; 3 use a method to maintain, same / stated, temperature ; 4 use a method to maintain, same / stated, light intensity ; 5 at each wind speed, measure / note / record / AW, the initial mass <u>and</u> final mass, in a fixed time ; 6 <i>idea of</i> equilibration / acclimatisation, of, plant / Japanese spiraea / apparatus ; 7 use at least three measurements for each wind speed <u>and</u> calculate a mean ; 8 safety comment with hazard <u>and</u> precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>Japanese spiraea / plant</td><td>irritant / allergy</td><td>gloves / mask / PPE</td></tr> <tr> <td>soil</td><td>biohazard / pathogens / allergy I irritant</td><td>gloves / mask / PPE</td></tr> <tr> <td>fan</td><td>hair / clothing / AW, trapped in fan</td><td>method to, control loose, hair / clothing / AW</td></tr> </tbody> </table> 9 method for measuring wind speed ; 10 <i>idea of</i> replacing water lost from plant (before each new measurement) ;	hazard	risk	precaution	Japanese spiraea / plant	irritant / allergy	gloves / mask / PPE	soil	biohazard / pathogens / allergy I irritant	gloves / mask / PPE	fan	hair / clothing / AW, trapped in fan	method to, control loose, hair / clothing / AW	6
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Question	Answer	Marks
1(a)(iii)	as wind speed, increases: rate of transpiration - increases or decrease in mass, in fixed / stated time - increases or rate of decrease in mass - increases or time taken, for a fixed decrease in mass - decreases ; ora	1
1(b)(i)	26.3 ;	1
1(b)(ii)	<i>any one from:</i> 1 use, same / stated, size of (blue) cobalt chloride paper ; 2 <i>idea of</i> use a (pink) colour standard (to determine end point) ; 3 <i>idea of</i> measure different positions on (lower) surface of leaf ; 4 <i>idea of</i> unbiased selection of, leaves / leaf position ;	1